

Lecturer

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Introduction to Econometrics

Lecture 1

Textbooks

- ❖ Enders, W. (2010), *Applied Econometric Time series Analysis*, 3rd Edition, John Willey & Sons: New York.
- ❖ Tsay, R. S. (2005), *Analysis of Financial Time Series*, 2nd Edition, John Willey and Sons: New York.

Lecture Objectives

- ❖ Understand what econometrics means.
- ❖ Differentiate between economic and econometric models.
- ❖ Know different types of data.
- ❖ Understand correlation vs causation

What is Econometrics?

❖ There does not exist a generally accepted answer to this question.

--Econometrics means “economic measurement”

Gujarati (2004) Basic Econometrics, 4th Edition. The McGraw-Hill Companies.

--Econometrics may be defined as the social science in which the tools of economic theory, mathematics, and statistical inference are applied to the analysis of economic phenomena.

Goldberger (1964) Econometric Theory, John Wiley&Sons, New York, p1.

What is Econometrics?

--Econometrics is the study of the application of statistical methods to the analysis of economic phenomena.

Kennedy (1998) *A guide to econometrics*. 4th Edition. The MIT Press Cambridge Massachusetts.

--Econometrics is based upon the development of statistical methods for estimating economic relationships, testing economic theories, and evaluating and implementing government and business policy.

Wooldridge (2002)

What is Econometrics?

preferred:

--Econometrics is the application of statistical and mathematical methods to the analysis of economic data, with a purpose of giving empirical content to economic theories and verifying them or refuting them.

Maddala (1992) Introduction to Econometrics. 2nd Edition. MacMillan Publishing Company.

Econometrics tests these theories

1. economic theory
2. economic stats
3. mathematics

Behavioural eqn

$$q = \alpha + \beta p + \varepsilon \quad \beta < 0$$

$q = Ap^\beta$ - inv rel. btwn price & quantity

$$\ln(q) = \ln(Ap^\beta)$$

$$\ln q = \ln A + \beta \ln p$$

The Aims of Econometrics

1. ❖ Formulation of econometric models i.e. formulation of economic models in an empirically testable form. *specification aspect of econometric modeling*
2. ❖ Estimation and testing of these models with observed data. *inference aspect of econometric work*
3. ❖ Use of these models for prediction and policy purposes.
Based on historical data, if model is good enough, you can use it to predict future

Lucas Critique: most regressions we use, we use hist. data, but when it comes to policymaking

Economic and Econometric Models

❖ Economic model

simplified rep. of real world process

Is a set of assumptions that approximately describes the behaviour of an economy (or a sector of an economy)

❖ Example

Quantity demanded of a commodity = $f(\text{price of the commodity, price of other commodities, tastes})$

Econometric model

❖ Consists of the following:

1. A set of behavioural equations derived from the economic model.
2. A statement of whether there are errors of observation in the observed variables.
3. A specification of the probability distribution of the disturbances (and errors of measurement)

Econometric model

ε should not be correlated with variables. If so, then unreliable
 $\hookrightarrow$ = endogeneity issue

❖ Example

(i) The behavioural equation: $q = \alpha + \beta p + \varepsilon$
DV IV

Where q is the quantity demanded and p is the price.

Here q and p are the observed variables *things that aren't observable go into ε*
 ε is the disturbance term.

(ii) A specification of the probability distribution of ε which says that $E(\varepsilon|p)=0$ and that the values of ε for the different observations are independently and normally distributed with mean zero and variance σ^2 .

problems

1. model over-simplified
2. we are making unrealistic assump.

Types of econometrics

❖ Econometrics may be divided into two broad categories:

(i) Theoretical econometrics

❖ Theoretical econometrics is concerned with the development of appropriate methods for measuring economic relationships specified by econometric models. *doesn't use data to run analysis*

➤ Leans heavily on mathematical statistics.

➤ It must spell out the assumptions of the method, its properties, and what happens to these properties when one or more of the assumptions of the method are not fulfilled.

i.e. rely on simulations

Types of econometrics

our approach for the rest of this course

uses data to do analysis
(ii) Applied econometrics

❖ Applied econometrics uses the tools of theoretical econometrics to study some special field(s) of economics and business.

➤ E.g. production function (economic growth models), investment function, demand and supply function, money demand function etc.

prod fn: $Y = AK^\alpha L^{1-\alpha}$
used to find economic growth

A - total factor productivity
L - labour

Data

❖ Economic data sets come in a variety of types

1. Cross-sectional data

- ❖ A cross-sectional data set consists of a sample of individuals, households, firms, cities, states, countries, or a variety of other units, taken at a given point in time.
- We can often assume that they have been obtained by random sampling from the underlying population.
- E.g. obtain information on wages, education, experience and other characteristics by randomly drawing 500 people from the working population. (use surveys for this).
- The analysis of cross-sectional data is closely aligned with the applied microeconomics fields e.g. labour economics, industrial organization, urban economics, demography, and health economics

- Data Set on Wages and Other Individual Characteristics

- obsno wage educ exper female married

- 1 3.10 11 2 1 0

- 2 3.24 12 22 1 1

- 3 3.00 11 2 0 0

- 4 6.00 8 44 0 1

- 5 5.30 12 7 0 1

-

-

-

- 525 11.56 16 5 0 1

- 526 3.50 14 5 1 0

2. Time series data

- ❖ A time series data set consists of observations on a variable or several variables over time.
- E.g. stock prices, MS, ^{money supply} CPI, GDP, Exchange rate, etc.
- Unlike the arrangement of cross-sectional data, the chronological ordering of observations in a time series matters.
- Some variables tend to display clear trends.
- Data frequency at which the data are collected include: daily, weekly, monthly, quarterly, and annually.

3. Panel or Longitudinal data

↳ increases data pts

-
- ❖ A panel data (or longitudinal data) set consists of a time series for each cross-sectional members in the data set.
 - The same cross-sectional units (individuals, firms, or countries) are followed over a given time period.
 - The ordering in the cross-section of a panel data set does not matter.

Descriptive statistics

- ❖ Mean
- ❖ Variance
- ❖ Skewness
- ❖ Kurtosis *behaviour of a tail*
- ❖ Maximum
- ❖ Minimum

Mean

- ❖ Is the first moment. *of random. var*
- ❖ Is the average value of the series.
- ❖ It measures the central location of the distribution
median is better - not sensitive to outliers (ie it's robust)

Variance

- ❖ It's the second moment.
- ❖ It measures the variability of a variable.
- ❖ The positive square root of variance is the standard deviation of a variable.

Skewness

+ve & -ve skewness

❖ Is the third moment.

❖ Is a measure of asymmetry of the distribution of the series around its mean.

+ve skewness of asset returns:

There were frequent small losses + few large gains

Kurtosis

tells about behaviors of a tail

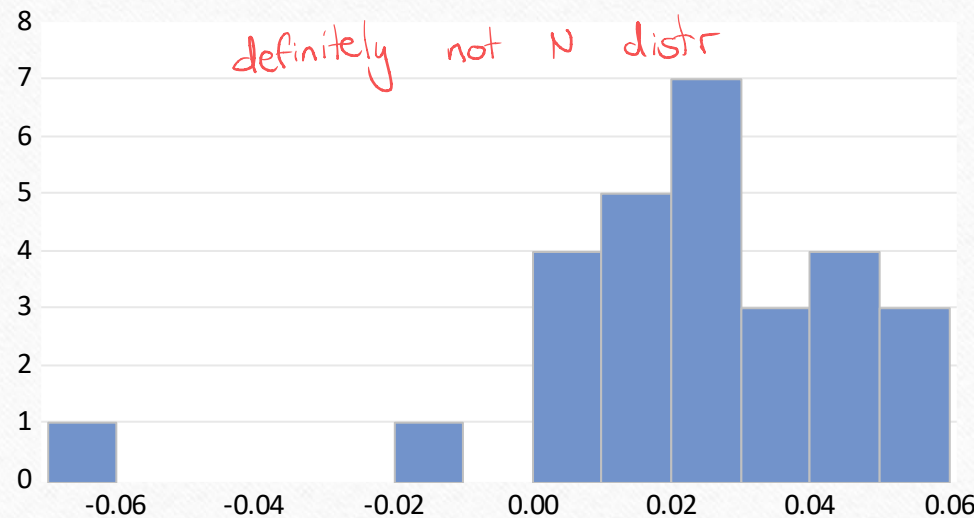
- ❖ It's the 4th moment.
- ❖ It measures the tail behaviour of a variable relative to the centre of the distribution.
- ❖ There are three types of kurtosis
 - Mesokurtic (normal distribution) $= 3$
 - Platykurtic (less than normal) < 3
 - Leptokurtic (more than normal) > 3

Max and Min

- ❖ Max (maximum) and Min (minimum) are the maximum and minimum values of the series in the current sample.

Descriptive statistics: DLGDP for SA

/lagged
1st diff $\rightarrow y_t - y_{t-1} = \Delta y$



Series: DRGDP	
Sample 1995 2023	
Observations 28	
Mean	0.022556
Median	0.025109
Maximum	0.054524
Minimum	-0.063674
Std. Dev.	0.023975
Skewness	-1.618758
Kurtosis	7.126548
Jarque-Bera	32.09490
Probability	0.000000

2,3%

in 2020 (Covid)

tells you whether it's N distr
sig $\therefore$ not N distr

- comment on some vars, not all

Correlation vs Causation

Correlation is not Causation

Establishing a causal relationship between key variables is essential for learning from empirical research (e.g. Interest rates and GDP). It is also an enormous challenge in social science.

Imagine variables x and y are correlated. There can be several reasons for this correlation, which are not mutually exclusive:

- ❖ Changes (or variation) in x drive changes (variation) in y : $x \Rightarrow y$
- ❖ Changes in y drive changes in x : $x \Leftarrow y$
- ❖ Correlated through a third variable: Changes in z drive changes x and y : $z \Rightarrow x$ and $z \Rightarrow y$ BUT $y \not\Rightarrow x$ nor $x \not\Rightarrow y$

That is, the cause underlying the correlation may be indirect and/or unknown

Correlation vs Causation

Causation

Causation can be established only when the explanatory variable is exogenous or when we can exploit exogenous variations of the endogenous explanatory variable!

Exogenous vs. Endogenous Variables

Economic Meaning:

Exogenous variables are determined outside of the model, e.g., age, sex, race.

Endogenous variables are determined within the model, i.e., as a simultaneous structural equations system.

Econometric Meaning:

An explanatory variable x is endogenous if it is correlated with the error terms.

If x is uncorrelated with the error terms, then x is said to be exogenous.

In a model like $y = x\beta + u$, x must be exogenous, i.e. $E(u|x) = 0$, for unbiased and consistent estimates of β . This assumption fails in the presence of measurement error, simultaneous equations and omitted variables in x .